

Sandipan Nath

Incoming PhD. Candidate in Computer Science

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Education

Technische Universität Graz , Computer Science PhD position focusing on integration of Spiking Neural Network (SNN) models into wireless communication systems.	Graz, Austria Expected 2026-09
Friedrich-Alexander-Universität Erlangen-Nürnberg , Data Science - Current grade of 1.3 on a scale of 1.0 (best) to 5.0 (worst) - Research focus on nonlinear dynamics, stochastic resonance, and reservoir computing - Master's thesis on structural heterogeneity in coupled FitzHugh-Nagumo neuron networks	Erlangen, Germany Apr 2024 – present
University of Alberta , Computing Science - GPA of 3.34 on a scale of 4.0 (best) to 0.0 (worst) - Murray Thomas Gibson Memorial Scholarship in Mathematics (2019) - University of Alberta Undergraduate Scholarship (2019) - Department of Mathematical Statistical Sciences Scholarship (2018)	Edmonton, Canada Sept 2018 – Nov 2022

Experience

Technische Universität Graz , University Project Assistant Staff member at the Institute of Machine Learning and Neural Computation.	Graz, Austria Starting 2026-09
Siemens Energy Global A.G. , Automation and Data Science Developer (Working Student) - Designed and implemented a centralized data management system to streamline data flow between HVDC system design tools - Built a consistent nomenclature system and supporting database from scratch using Python and MATLAB - Developed an interactive dashboard UI to manage tool interoperability via the database 15 hours/week	Erlangen, Germany Jan 2025 – July 2026 1 year 7 months
GaoTek Inc. , Software Development Intern - Participated in all stages of software development, including design, coding, and testing - Created and ran automation scripts using open-source tools - Collaborated with product and design teams using AGILE methodology 20 hours/week	Toronto, Canada May 2023 – Sept 2023 5 months
University of Alberta , University Teaching Assistant - Supervised practical labs and marked assignments/tests - Provided coursework assistance to undergraduate students 8 hours/week	Edmonton, Canada Jan 2021 – Apr 2021 4 months

Projects

Reinforcement Learning for Control Theory - Built a world model using LSTM for complex system dynamics - Trained a controller to balance a ball on an arc based on learned dynamics - Skills: Python, Reinforcement Learning, LSTM, MDN, Control Theory
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Forecastability in Chaotic Neural Dynamics

- Built an Echo State Network to model Hindmarsh–Rose neurons in spiking, periodic bursting, and chaotic states
- Explored the network’s ability to forecast nonlinear and chaotic time series
- Skills: Reservoir Computing, ESN, Nonlinear Dynamics

Object Detection for Arctic Wildlife

- Implemented YOLO-based object detection for caribou localization in camera trap images
- Evaluated performance in Arctic conditions
- Skills: Python, Computer Vision, PyTorch, YOLO

Modelling Drug Effects in Tumor Cell Dynamics

- Fitted ODE models to describe cell growth, drug availability, and cell killing mechanisms
- Simulated and predicted tumor growth responses to drug regimens
- Skills: ODE Modeling, Systems Biology, SciPy, Numerical Methods

Skills

Programming Languages

Machine Learning & AI

High-Performance Computing

Data & Databases

Tools & Platforms

Languages

English: C2 (Proficient/Native)

Hindi: C2 (Proficient/Native)

Bengali: Native

German: B1.1 (Intermediate)

Certificates

GRE General Test

Mar 2024

- Verbal Reasoning: 161/170, Quantitative Reasoning: 162/170, Analytical Writing: 3.5/6

TOEFL iBT

Jan 2024

- Total Score: 109/120 (Reading: 28, Listening: 30, Speaking: 23, Writing: 28)